L.S.F

LAU
School of Engineering

ELE 430, Signals and Systems
Exam 2, Spring 2015
Instructor: Dr. Chadi Abou-Rjeily

10452

Problem 1: (12 points: (3 × 4))

a. Starting from the definition of the inverse Fourier transform (IFT), find the IFT of $\delta(f - f_0)$.

Consider the signal $x(t)$ given by:

$$x(t) = \sin(2\pi f_0 t)$$

b. Deduce from part (a) the Fourier transform $X(f)$ of $x(t)$.

c. Plot the magnitude spectrum and the phase spectrum.

d. Show that $X(f)$ is odd.

Problem 2: (13 points: (4, 1, 4, 4))

a. Starting from the definition of the Fourier transform (FT), find the FT of $\text{rect}\left(\frac{t}{w}\right)$.

b. Deduce from part (a) the FT of $\text{sinc}(t)$.

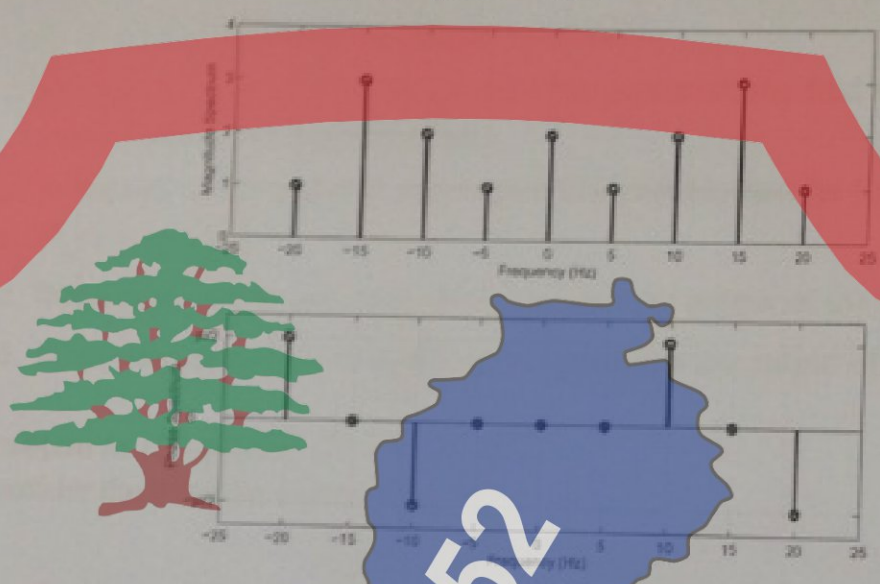
Consider the signals $x(t)$ and $y(t)$ related to each other by the relation: $y(t) = x(t)\cos(2\pi f_c t)$.

c. Using the convolution-multiplication duality, relate $Y(f)$, the FT of y(t), to $X(f)$, the FT of $x(t)$.

d. Provide a labeled sketch of $Y(f)$ when $x(t) = \text{sinc}(t)$ and $f_c = 5$ Hz.

Problem 3: (10 points: (2,4,2,2))

The magnitude and phase spectra of a periodic signal $x(t)$ are shown in the figure below:



a. Find the fundamental period of $x(t)$.

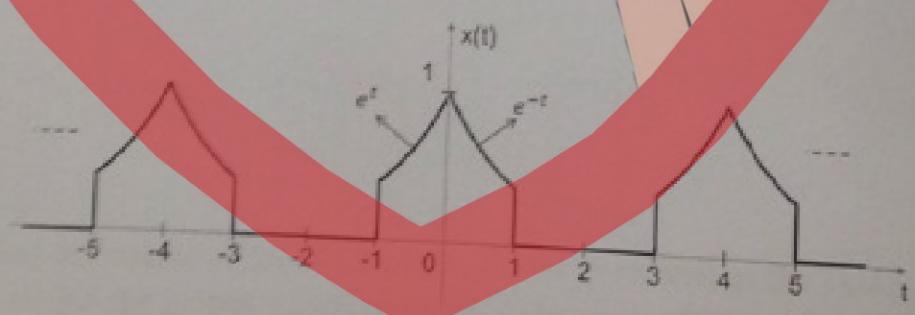
b. Is the signal real or complex? Justify your answer.

c. What is the average value of the signal?

d. Find the power of the second harmonic.

Problem 4: (10 points)

Consider the signal shown in the figure below:



What is the DC power of this signal?

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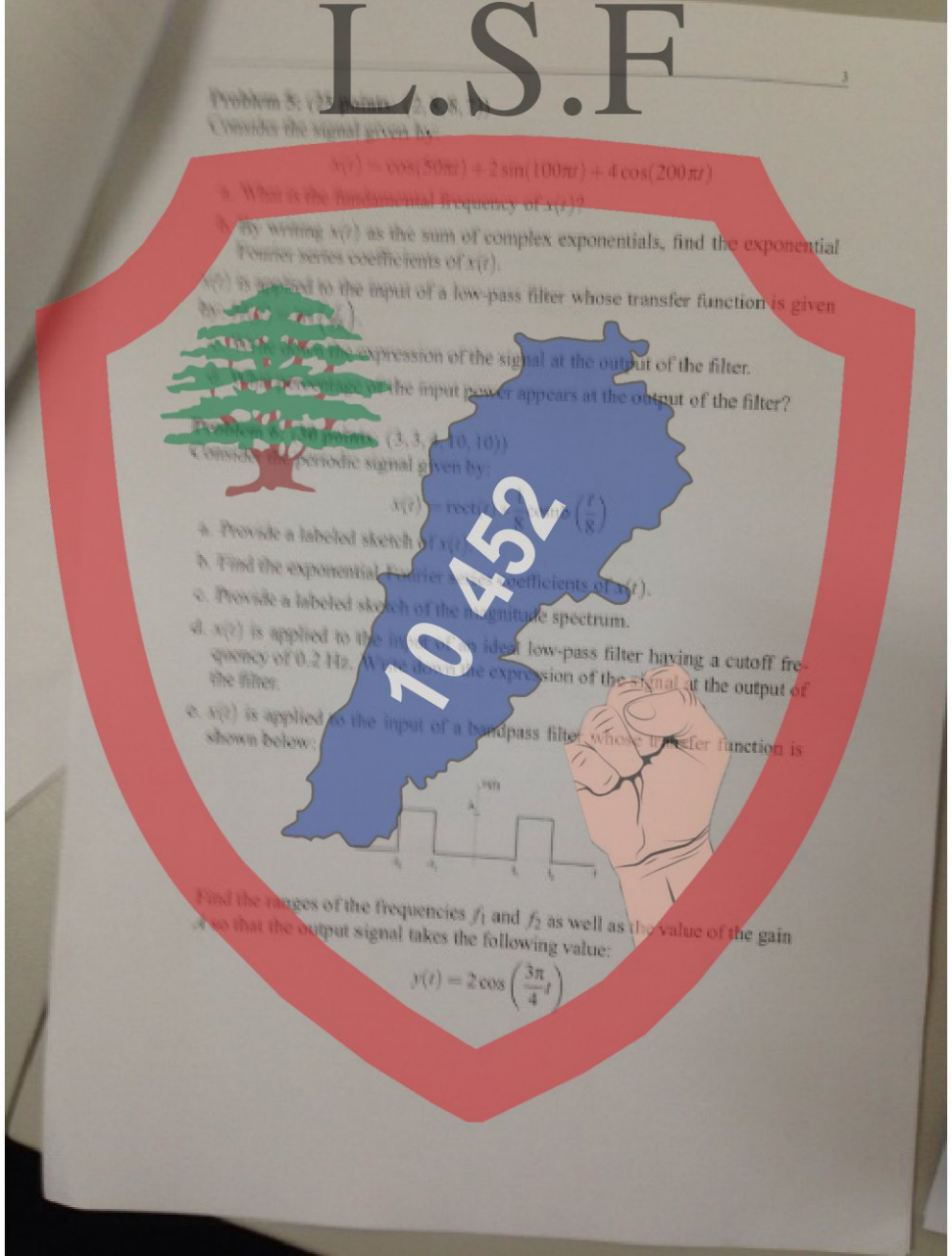
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